

Mayank Singh

SOFTWARE DEVELOPER

Details

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TECHNICAL SKILLS

Java
C++
Python
JavaScript
SQL
HTML5
CSS3
Spring Boot
React.js
Hibernate
Node.js
Maven
Git
Jenkins
MySQL
Oracle
MongoDB
PostgreSQL
Object-Oriented Programming
REST APIs
Agile
Microservices
Design Patterns
MS Office
CAD/CAM
MatLab

EDUCATION

Master of Science in Computer Science, Pace University, New York, NY, New York

MAY 2025

CS 505 (Introduction to Java), CS 604 (Computer System And Concepts), CS 608 (Algorithms and Computing Theory), CS 610 (Introduction to Parallel Computing), CS 612 (Concepts And Structures in Internet Computing), CS 623 (Database Management Systems), CS 627 (Artificial Intelligence), CS 629 (Computer Graphics), CS 632Q (Introduction to Natural Language Processing), CS 639 (Mobile Application Development), CS 691 (Capstone Project).

Bachelor of Technology in Mechanical Engineering, Manipal University Jaipur, India, Jaipur

MAY 2019

ACADEMIC PROJECTS

Supermarket Theft Detection System – Capstone Project (CS691)

Led AI integration for detecting suspicious behavior in CCTV footage using Python, TensorFlow, and OpenCV. Developed WebSocket-enabled Django UI for live monitoring and alerts. Implemented machine learning pipeline with MobileNet and custom detection logic for real-time analysis. Managed team coordination, sprint retrospectives, and deliverables via Agile methodology.

Full-Stack Inventory Management System – CS610

Designed and developed a full-stack web application using Spring Boot and React.js. Implemented RESTful APIs for CRUD operations and integrated MySQL database. Applied MVC architecture, authentication via Spring Security, and unit tests using JUnit.

Database Optimization and Analytics – CS608

Designed normalized relational database schemas and optimized SQL queries for performance. Applied indexing and query tuning strategies for efficient transactional handling. Developed stored procedures and triggers to automate data integrity validation.

WORK EXPERIENCE

Teaching Assistant at Pace University, New York

JANUARY 2025 – MAY 2025

- Evaluated MS in CS Capstone Projects, reviewing code quality, system design, and documentation.
- Mentored students on software architecture, version control, and best practices in full-stack development.

Environment: Java, Spring Boot, React, AWS, Oracle, JUnit, Jenkins, Windows.

Software Developer at Manupatra Information Solutions, Noida, India

JANUARY 2020 – DECEMBER 2022

- Designed and maintained Spring Boot microservices for content ingestion, metadata enrichment, and search indexing using Elasticsearch.
- Developed file-automation scripts (Regex + Node.js) to classify and tag 100K+ legal documents, improving indexing speed by 45%.
- Created a multi-threaded parser to extract structured metadata from large unstructured text files.
- Implemented REST APIs for user authentication, notifications, and analytics dashboards using Spring Boot + Hibernate.
- Deployed and monitored services via Jenkins CI/CD, AppDynamics, and Docker, achieving 99.8% uptime.
- Collaborated in Agile sprints with cross-functional teams using Jira, participated in design reviews, and implemented JUnit + RestAssured test suites for regression automation.